

2.15 Cancer

▼ What is a tumour?

- Any kind of **mass-forming lesion** (lesion meaning an area which has **suffered injury from disease or injury**)
- It can be **neoplastic or non-neoplastic**
 - i.e they may or may not have escaped normal constraints on cell proliferation

▼ What is a neoplasm?

- The **autonomous growth of tissue** that have **escaped normal constraints on cell proliferation**
 - i.e any tissue that has grown too big without external stimulation, but with cell cycle defects
- **Cancers are malignant neoplasms**

▼ What is the difference between a benign and malignant neoplasm?

- **Neoplasia** describes cells that have **escaped normal constraints on cell proliferation**
- A **benign neoplasm** is **localised** and so is **harmless**
- A **malignant neoplasm** can **metastasise and invade other tissue locally/distantly** and so is dangerous

▼ How would you distinguish between a benign and malignant tumour upon physical examination?

- A **benign neoplasm** can be **moved around on examination** when palpating its surface
- A **malignant neoplasm** **cannot be moved around on examination** when palpating its surface
 - This is because it has often **invaded other tissue and become attached to it (i.e muscles)**

▼ What is metastasis?

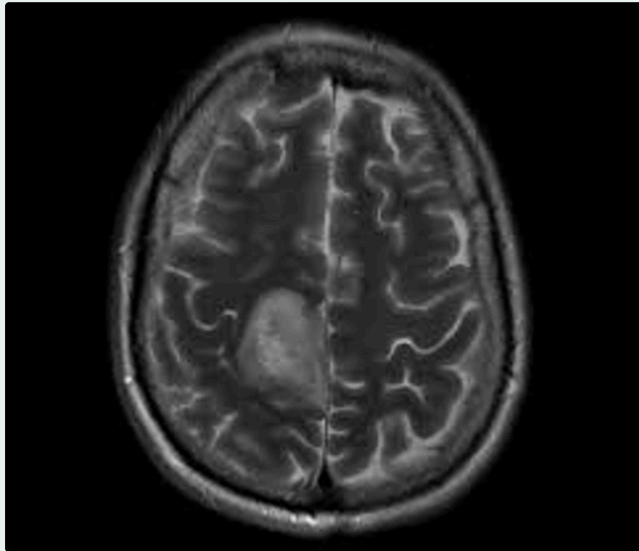
- This is when a **malignancy has spread to distant sites** around the body **via blood vessels**
- It is good to **diagnose a cancer before metastasis**

▼ Which malignant tumour is typically known to not always result in death?

- **Skin cancers (melanomas)**

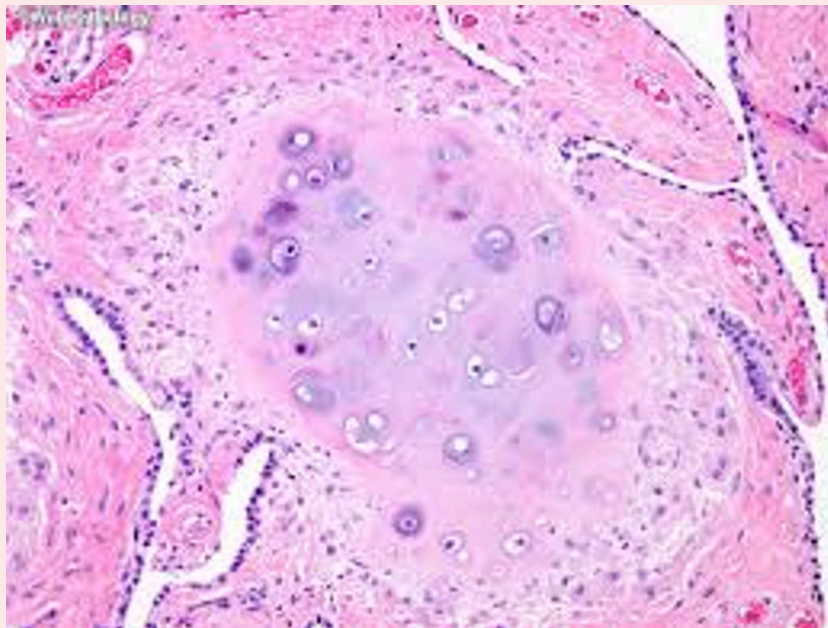
▼ Which benign tumour is typically known to result in death and why?

- **Brain tumours**
 - This is because it stops the flow of CSF between ventricles
- **Thyroid** and other **endocrine gland benign tumours** can also be harmful



▼ What is a hamartoma?

- A **localised, benign overgrowth** of one or more **mature cell types** - they are **harmless**
- A **discrete mass (tumour)** is **formed** within this tissue that **consists of multiple cell types**
- It is **not simply a "benign neoplasm"** because a **hamartoma** is distinguished by **having multiple cell types**
 - The cell types are usually already present in the tissue, they just tend to **clump together to form this distinct mass**



▼ Is a hamartoma neoplastic?

- **Yes.**
- The **cells involved have escaped the normal constraints of cell proliferation** and this is how they combine together to form their distinct mass.

▼ What is a heterotopia?

- This is where there is a **normal group of cells (tissue)** being **found in parts of the body where they are not normally present**
 - i.e **pancreas tissue found in large intestine wall**

▼ What does the prefix "chondro" suggest in cancer classification?

- The tumour is **derived from cartilage**

▼ What does the suffix "-oma" suggest in cancer classification?

- A **benign tumour**

- ▼ What does the suffix "-sarcoma" suggest in cancer classification?
 - A **malignant tumour**
 - In any tumour classification, the **prefix** refers to the **type of tissue affected**, and the **suffix** refers to the **malignancy of the tumour**.
- ▼ What is the name given to a benign tumour of squamous epithelial cells?
 - A **squamous papilloma**
- ▼ What is the name given to a malignant tumour of squamous epithelial cells?
 - **Squamous cell carcinoma**
- ▼ Which tissues can a squamous cell neoplasm affect?
 - **Skin**
 - **Oesophagus**
 - **Cervix**
- ▼ What is the name given to a benign tumour of glandular epithelial cells?
 - **Adenoma**
- ▼ What is the name given to a malignant tumour of glandular epithelial cells?
 - **Adenocarcinoma**
- ▼ Which tissues can a glandular cell neoplasm affect?
 - **Breast**
 - **Colon**
 - **Pancreas**
 - **Thyroid**
- ▼ What is the name given to a benign tumour of transitional epithelial cells?
 - **Transitional papilloma**
- ▼ What is the name given to a malignant tumour of transitional epithelial cells?
 - **Transitional cell carcinoma**
- ▼ Which tissues can a transitional cell neoplasm affect?
 - **Bladder**
- ▼ What is a transitional cell?
 - A **cell that varies in shape depending on whether the tissue is being stretched**.
- ▼ What is the name given to a benign tumour of smooth muscle cells?
 - **Leiomyoma**

▼ What is the name given to a malignant tumour of smooth muscle cells?

- **Leiomyosarcoma**

▼ Which structures can a smooth muscle cell neoplasm affect?

- Uterus
- Colon
 - **Contractile tissues**

▼ What is the name given to a benign tumour of bone cells?

- **Osteoma**

▼ What is the name given to a malignant tumour of bone cells?

- **Osteosarcoma (osteogenic sarcoma)**

▼ Which structures can a bone neoplasm affect?

- Arm
- Leg

▼ What is a connective tissue (soft tissue) cancer called?

- **Sarcomas**

▼ What is an epithelial tissue cancer called?

- **Carcinomas**

▼ What are the two types of haematological neoplasms?

- **Lymphomas**
- **Leukaemias**
- **Benign tumours are very rare for haematological neoplasms.**

▼ What is the difference between a lymphoma and a leukaemia?

- A **lymphoma** is a cancer **derived from lymphocytes**
 - The lymphocytes would be found anywhere in the bloodstream, not the bone marrow
- A **leukaemia** is a cancer **derived from the bone marrow**
- However, since **lymphocytes that arise from the bone marrow**, what are these cancers called?
 - Cancers in the **lymphocytes still present in the bone marrow** are classified as **lymphocytic leukaemias**

▼ What is a teratoma?

- A **tumour derived from a germ cell**
- Can be **derived from all three germ cell layers** (i.e endoderm, mesoderm, ectoderm)
- They can occur in **immature or mature cells**.

▼ Where are the most common sites that you would find teratomas at?

- **Testes**
- **Ovaries**

▼ Teratomas can be derived from both immature and mature cells. Which is worse?

- **Immature teratomas are worse** because **immature cells** have much **higher proliferative capacity**
 - This leads to higher risk of malignancy and hence systemic damage

▼ Recall four examples of tumours with the suffix -oma but still being malignant.

- **Lymphoma** (lymphocytes)
- **Melanoma** (skin)
- **Hepatoma** (liver cells)
- **Teratoma** (germ cells)

▼ Recall four ways that benign and malignant neoplasms differ from each other.

- Extent of **invasion**
- Malignancy
- Differentiation
- Growth pattern

▼ Which of these differences can be used to initially identify whether a growth is benign or malignant in a lab?

- **Differentiation**
- **Growth pattern**

▼ What is the difference between dysplasia and cancer?

- **Cancer usually involves invasion**, dysplasia does not
 - **Invasion** is seen in **malignant neoplasms**, **dysplasia** is seen in **benign neoplasms**
- **Invasion (in cancer)** involves **cells directly invading adjacent connective tissue** and **other structures** (i.e blood vessels)
- **Dysplasia** means the **cells show cytological features of cancer** (i.e ratio of cytoplasm:nucleus) but **do not invade**
 - i.e the **intermediate stage** between normality and cancer

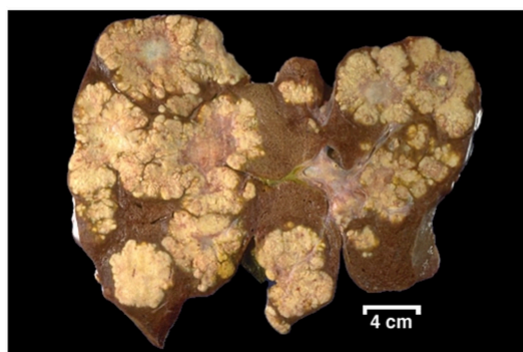
▼ What is the difference between metastasis of benign and malignant tumours?

- **Benign tumours are non-invasive**, **malignant tumours are invasive**
 - **NOTE:** Not all malignant tumours metastasise (spread through the blood). Some simply invade locally.

▼ What is the difference between invasion and metastasis?

- **Invasion** describes **local encroachments of a neoplasm**
 - i.e **breast cancer attaching to pectoralis major muscle**
- **Metastasis** describes **systemic encroachments of a neoplasm** where the **neoplasm spreads through the blood to affect other structures**

Liver metastases from a patient with pancreatic cancer



▼ What is meant by differentiation of tumours?

- **Differentiation** describes how much the cells of the tumour resemble the cells they originated from
- **Well differentiated** cells look a lot like the cells they derive from
- **Poorly differentiated** cells do not really look like the cells they derive from

▼ What is the difference between malignant and benign tumour cells cytologically?

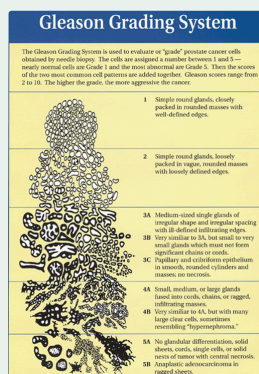
- **Malignant tumour cells** tend to have a much **higher nuclear:cytoplasmic ratio** than the cells of the tissue they are derived from
- **Benign tumour cells** don't typically possess this trait.

▼ What is meant by the growth pattern of tumours?

- **Growth pattern** describes how well the architecture of the neoplastic tissue resembles the architecture of the tissue it is derived from
 - It is similar to differentiation, but whilst **differentiation** refers to **cytological appearance**, **growth pattern** is all about **tissue architecture**

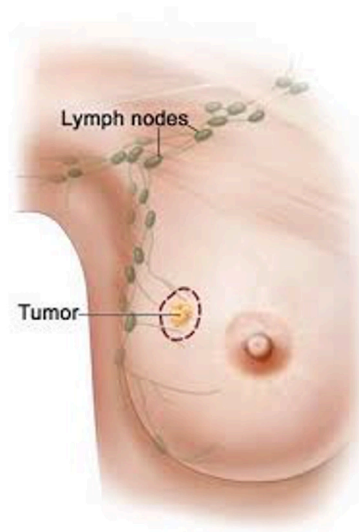
▼ What is the difference in growth patterns between tumour cells and their derived cell types?

- **Poorly differentiated tumours** look a lot **less similar** to the tissue they derived from
- **Well differentiated tumours** look a lot **more similar** to the tissue they derived from



- ▼ Recall five ways that neoplasms can spread.
 - **Direct extension** (structures being in close proximity to each other)
 - i.e **thyroid cancer patients having coarse voices** due to **proximity of recurrent laryngeal nerve to thyroid gland**
 - **Haematogenous** (via blood vessels)
 - **Lymphatic** (through lymphatic vessels to lymph nodes)
 - **Transcoelomic** (via seeding of body cavities such as pleural cavities)
 - **Perineural** (via nerves)
 - Through injections (i.e biospies)
- ▼ Which types of blood vessels are typically invaded in metastases and why?
 - **Veins and capillaries**
 - This is because **they have a lot less smooth muscle** and so **their walls are much thinner**
- ▼ What is the function of the thoracic duct?
 - To **drain fluid from the lymph nodes back into the circulation**
- ▼ Which type of cancer typically metastasises through the lymphatics?
 - **Epithelial cell** cancers
- ▼ Image showing how lymphatic spread of breast cancer occurs through the lymphatic capillaries and axillary nodes.

lymphatic spread to axillary nodes from breast cancer



- ▼ What is the 'TNM' system used for?
 - To **describe the stage** and **how much a tumour has spread**

- ▼ Describe how the TNM system works.
- **T = size of tumour and extent of invasion by the tumour**
 - **N = number of lymph nodes affected by the tumour**
 - **M = presence of distant metastases (how bad the metastasis is)**